Nicholas Belev

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EDUCATION

McGill University, Montreal, QC — Graduated: May 2026
B.Sc. Computer Science & Artificial Intelligence; Minors in Mathematics, Finance
GPA: **3.92 / 4.00**
Relevant Coursework: Machine Learning, Algorithms, Statistics, Quantitative Risk Management, Portfolio Theory.

EXPERIENCE

SS&C Algorithmics, *Associate Software Engineer* — Toronto, ON — Jul 2026 – Present
- Develop **ARIA**, SS&C's AI agent system that handles complex financial analytics requests (portfolio analysis, scenarios, simulations); contribute to its Java middleware and AWS cloud infrastructure.
- Built **AI cost tracking and enforcement** on AWS Lambda, monitoring LLM usage per client and enforcing client spending limits.
- Extended ARIA for **multi-tenant, multi-client** deployment and improved its Portfolio Insights agent's analysis of holdings, exposures, and risk statistics.
- Implemented **RAG-based persistent memory** so ARIA retains client preferences, project context, and relevant history across conversations.

Aspequity, *Co-Founder & Chief Technology Officer* — Boston, MA — Sep 2025 – Present
- Co-founded a private equity forecasting software business serving **paying institutional clients**; own development of its C++ cash flow and valuation modeling platform.
- Built a pipeline to ingest private equity fund reports into structured historical data, feeding **regression-based forecasts** of fund contributions, distributions, and NAV.
- Extended the **Yale cash flow modeling framework** with historical fund data and regression-based estimation; developed PostgreSQL infrastructure and work toward secure client API access.

Santander Bank, *Data Analytics Intern* — Boston, MA — Jun 2025 – Aug 2025
- Led the firm's first **LLM audit**, designing adversarial prompts, automated API tests, and statistical metrics to assess model risk, prompt injection defenses, and PII handling.
- Built a **Power BI dashboard** using Oracle SQL and DAX to forecast audit workloads and guide staffing across a **150+ person** department.
- Refactored an AML text-matching tool by optimizing TF–IDF vectorization and fuzzy matching, delivering a **3x runtime improvement**.

Northfield, *Quantitative Analyst & Software Engineer* — Boston, MA — May 2024 – Sep 2024
- Developed Python portfolio construction and optimization analytics supporting long/short equity strategies for a **$65B AUM** asset manager.
- Built SQL/Python pipelines for factor attribution, volatility monitoring, and portfolio performance analysis.
- Migrated **16,000+** legacy factor model files to PostgreSQL and built a FastAPI service enabling **sub-10 ms** risk queries for portfolio managers.

PROJECTS

Othello AI Agent Competition (GitHub) — Oct 2024 – Dec 2024
- Engineered a competitive game-playing AI using alpha–beta pruning, iterative deepening, and Zobrist hashing under **2s** move limits and **500MB** memory constraints.
- Placed **top 10 of 200 teams**; achieved **100%** wins vs. greedy agents and **99%** vs. stochastic agents.

Custom ETF Portfolio, *Team Lead* — Sep 2025 – Dec 2025
- Led a **6-person team** to build a factor-aware equity portfolio, earning the **highest overall project evaluation**.
- Analyzed **10+ years** of returns using Fama–French factors and rolling OLS regressions; constrained exposures to reduce idiosyncratic risk to **under 20%**.

Financial Risk Modeling & VaR Forecasting (GitHub) — Mar 2026 – Apr 2026
- Modeled ETF returns using **ARMA-GARCH(1,1)** with Student-t innovations to capture volatility clustering and heavy tails.
- Forecasted **95%/99% VaR and Expected Shortfall** using historical simulation, EVT, and parametric methods; evaluated forecasts through rolling backtests.

TECHNICAL SKILLS

Languages: Java, Python, C++, C, SQL
Frameworks & Libraries: Spring Boot, Spring AI, LangChain4j, FastAPI, Pandas, NumPy, scikit-learn
Cloud & Tools: AWS (Lambda, EC2), PostgreSQL, Docker, Git, Linux, Power BI